



Artificial Intelligence in Medicine:

Foundations and Clinical Applications

Instructor



Sara El-Sayed El-Metwally

Associate Professor of Computer Science,
Computer Science Department,
Faculty of Computers and Information,
Mansoura University



<https://sarael-metwally.github.io/SaraEl-MetwallyLab/>



sarah_almetwally4@mans.edu.eg



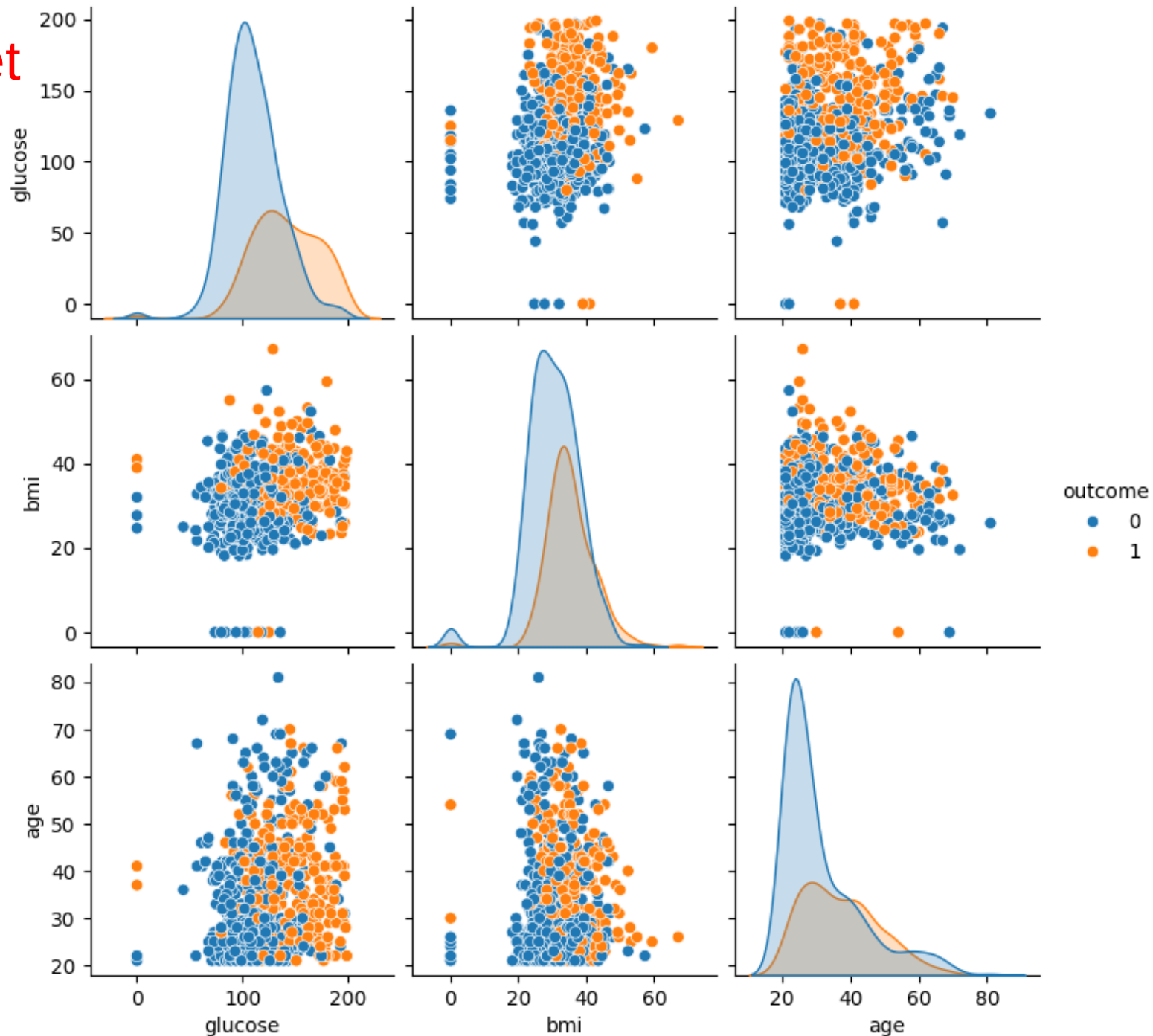
<https://www.facebook.com/sara.almetwally>

Case Study: Diabetes Dataset

```
import seaborn as sns
```

```
sns.pairplot(data, hue="outcome", vars=["glucose", "bmi", "age"])  
plt.show()
```

- A **pairplot** is a data visualization method that plots pairwise relationships between all numeric variables in a dataset. It generates an $N \times N$ grid of plots: off-diagonals show scatter plots between different variables to highlight correlations, while the diagonal shows histograms (or KDEs) displaying the distribution of individual variables.
- The **hue** parameter in a Seaborn pairplot colors the data points based on a specific categorical variable.

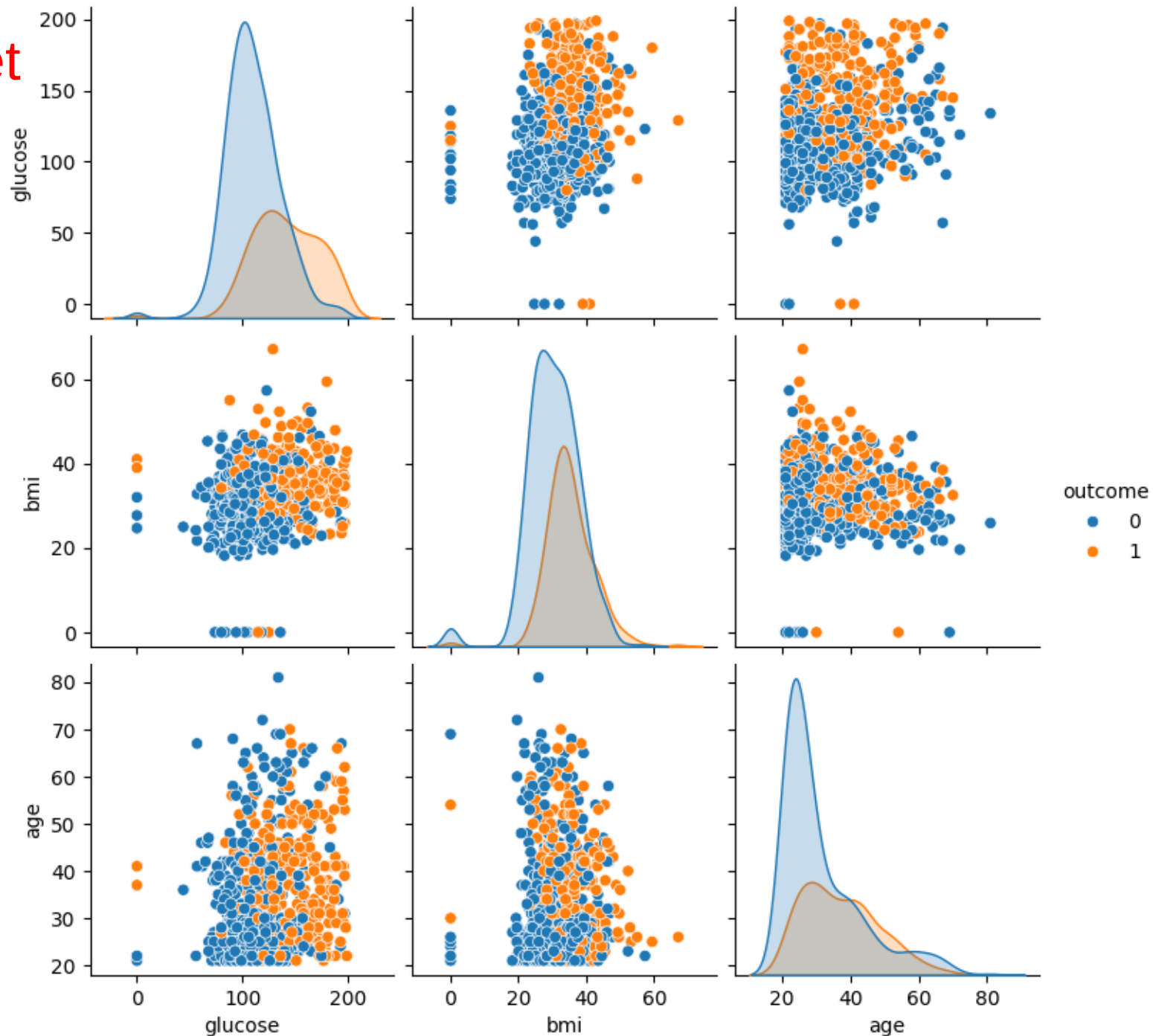


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- **Diagonal Curves:**
- **X-axis:** the value of the variable (e.g., glucose level)
- **Y-axis (height of curve):** roughly, how many patients have values around that point, taller = more patients cluster there.
- **Blue curve** = distribution of that variable only among patients without diabetes (outcome=0).
- **Orange curve** = distribution of that variable only among patients with diabetes (outcome=1).



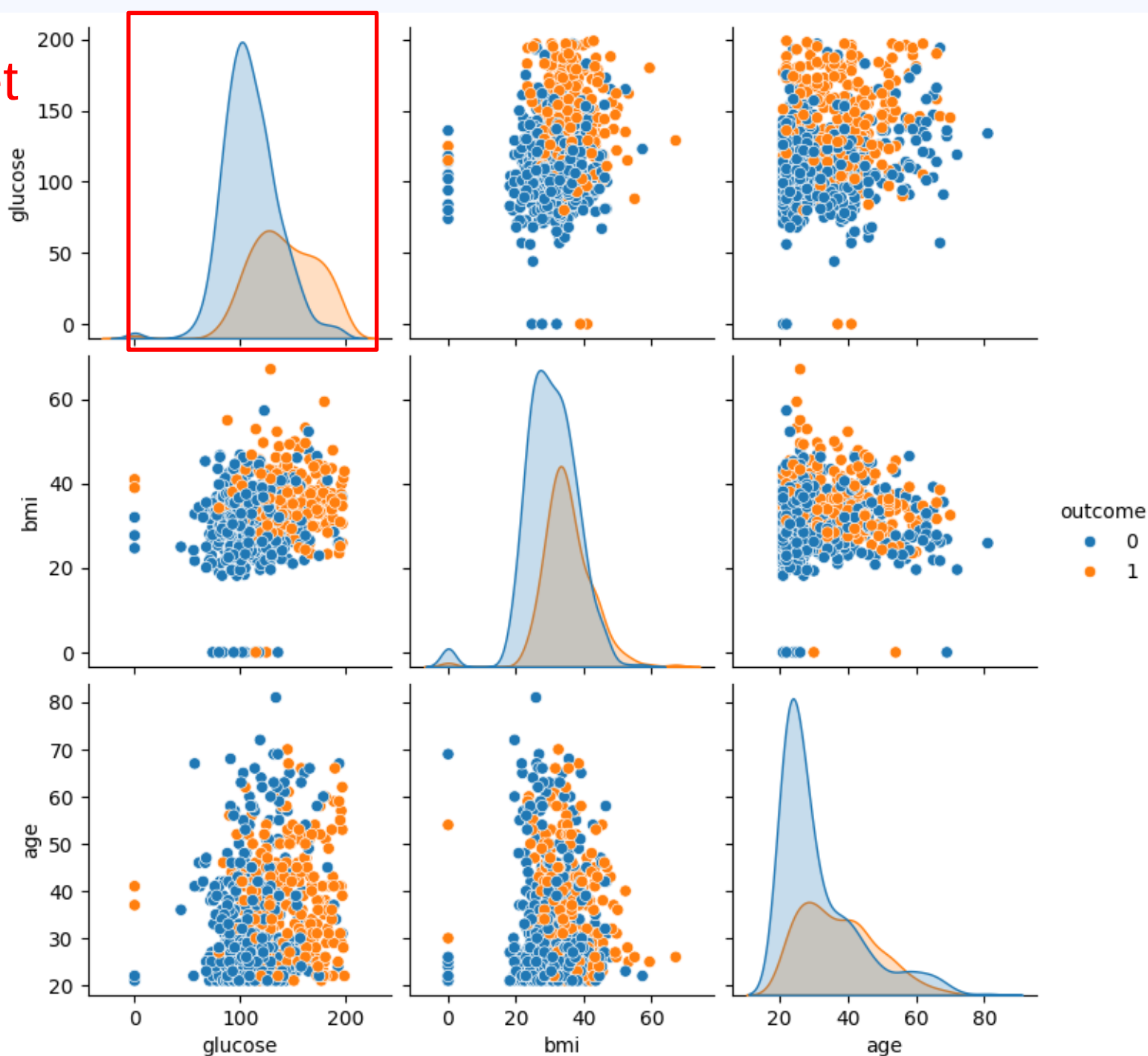
Case Study: Diabetes Dataset

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How to read the glucose curve specifically (top-left)

- The **blue curve** peaks around glucose~100-110 and is fairly narrow/tall, most non-diabetic patients cluster tightly around normal glucose.
- The **orange curve** is shifted right (peaking somewhere around 120-140) and is flatter/wider, diabetic patients' glucose values are both higher on average and more spread out.
- Where the curves overlap (shared shaded region) = the "ambiguous zone, glucose values common to both groups, where you genuinely can't tell which group a patient belongs to just from glucose alone.

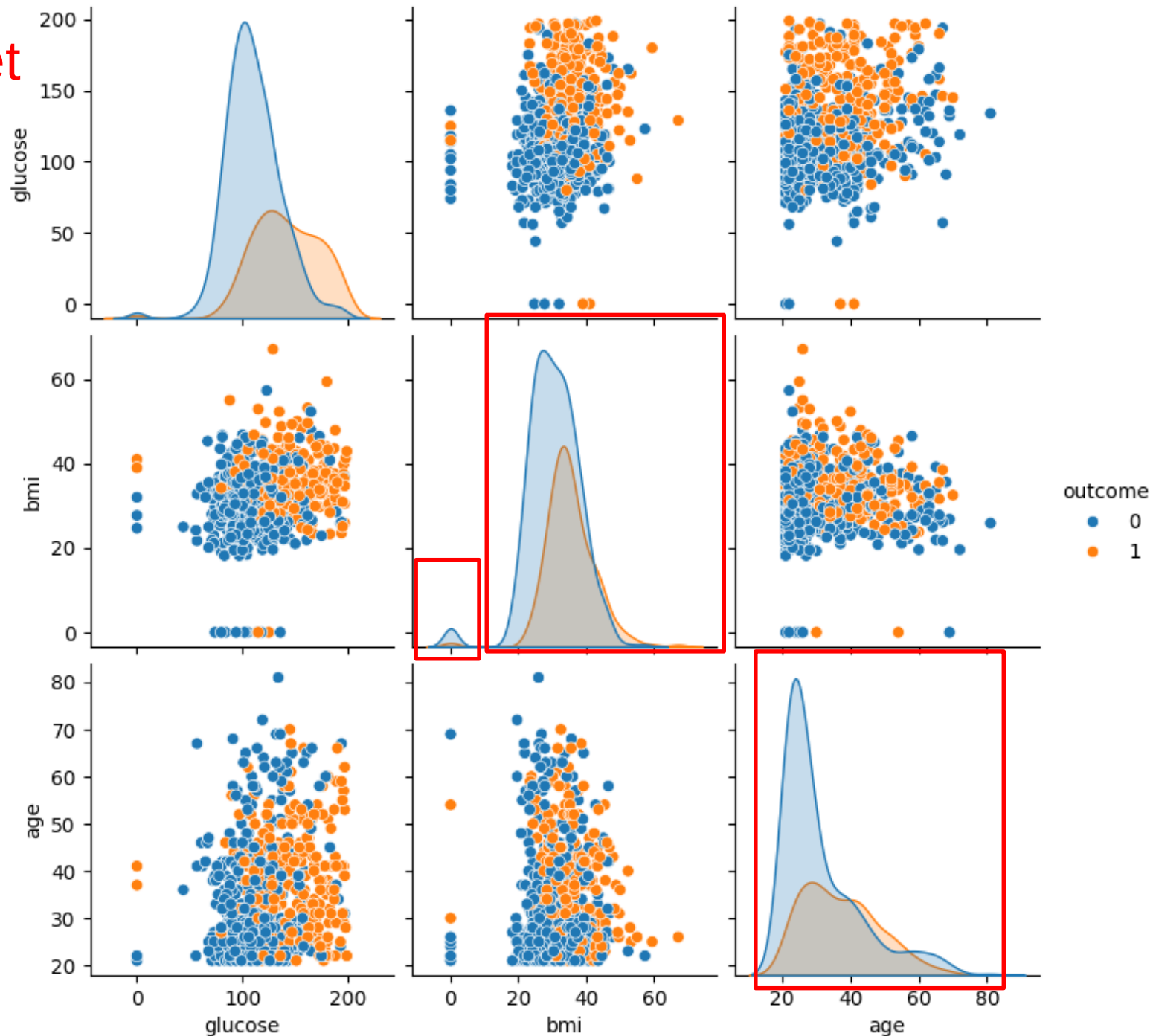


Case Study: Diabetes Dataset

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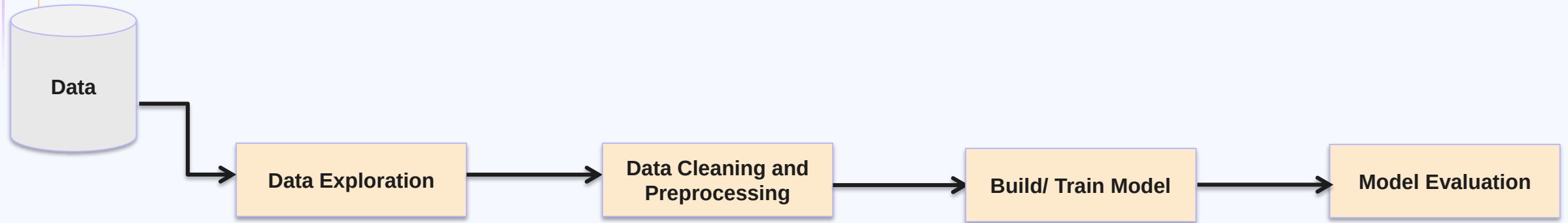
- If the two curves completely overlapped (identical shape and position), what would that tell you about that variable's usefulness for predicting diabetes
- If you had to rank glucose, BMI, and age by how well they separate diabetic vs. non-diabetic patients just from these three curves, what order would you put them in?
- Link:
<https://drive.google.com/drive/folders/1e53PXsrjzrhBc3-ksb7SsZyHFSzMssx?usp=sharing>



AI Workflow

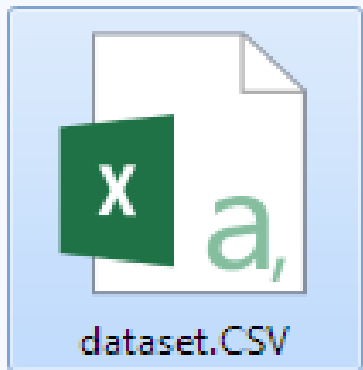


AI Workflow

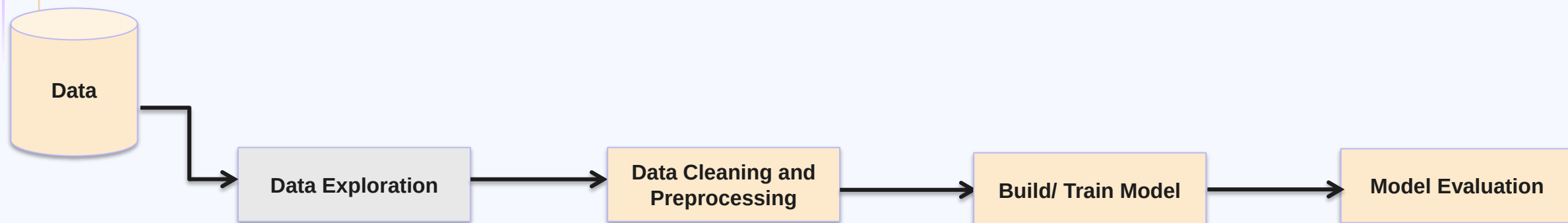


Dataset Overview

- **Observations:** 768 patient records (500 negative, 258 positive)
- **Target Variable:** Outcome (1 indicates diabetes, 0 indicates no diabetes)
- **Predictor Variables:** There are 8 features for each patient:
 1. **Pregnancies:** Number of times pregnant
 2. **Glucose:** Plasma glucose concentration from an oral glucose tolerance test
 3. **BloodPressure:** Diastolic blood pressure in mm Hg
 4. **SkinThickness:** Triceps skin fold thickness in mm
 5. **Insulin:** 2-Hour serum insulin in mu U/ml
 6. **BMI:** Body mass index (weight in kg / height in m²)
 7. **DiabetesPedigreeFunction:** Function indicating the likelihood of diabetes based on family history
 8. **Age:** Age in years

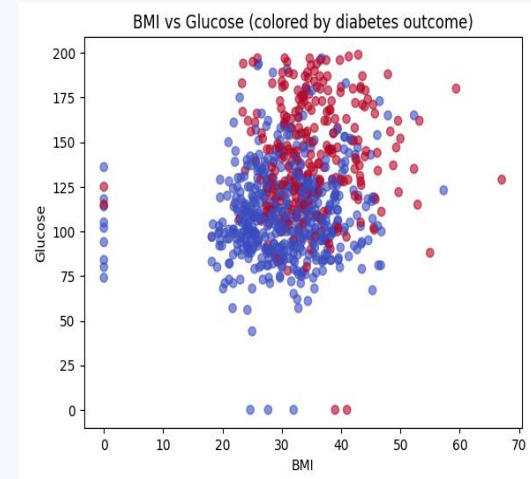
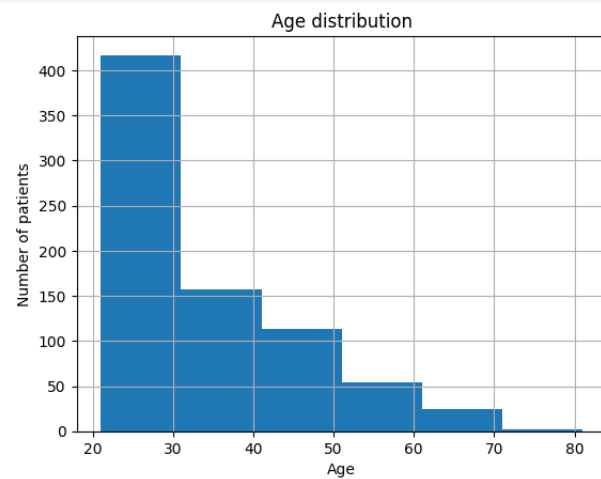


AI Workflow

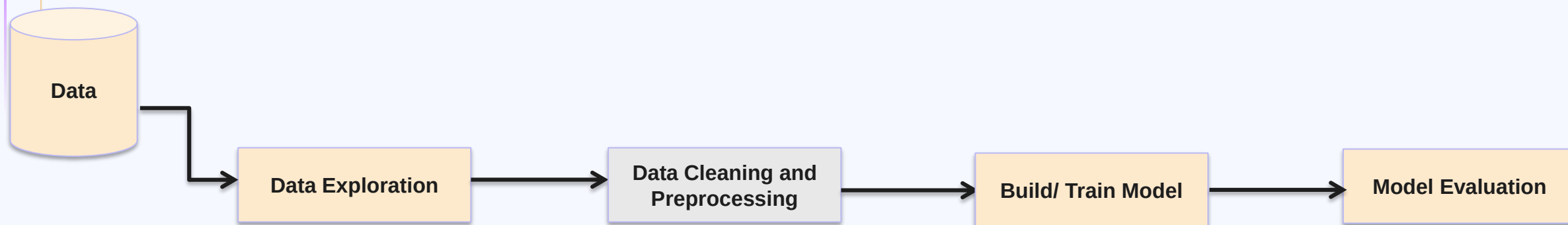


```
data.describe() # mean, min, max, etc. for each column
```

	pregnancies	glucose	blood_pressure	skin_thickness	insulin	bmi	diabetes_pedigree	age	outcome
count	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000
mean	3.845052	120.894531	69.105469	20.536458	79.799479	31.992578	0.471876	33.240885	0.348958
std	3.369578	31.972618	19.355807	15.952218	115.244002	7.884160	0.331329	11.760232	0.476951
min	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.078000	21.000000	0.000000
25%	1.000000	99.000000	62.000000	0.000000	0.000000	27.300000	0.243750	24.000000	0.000000
50%	3.000000	117.000000	72.000000	23.000000	30.500000	32.000000	0.372500	29.000000	0.000000
75%	6.000000	140.250000	80.000000	32.000000	127.250000	36.600000	0.626250	41.000000	1.000000
max	17.000000	199.000000	122.000000	99.000000	846.000000	67.100000	2.420000	81.000000	1.000000

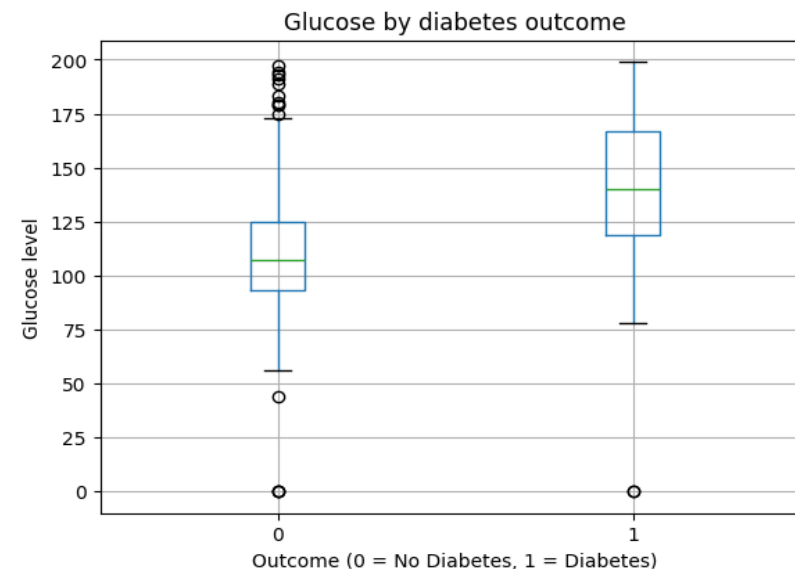


AI Workflow



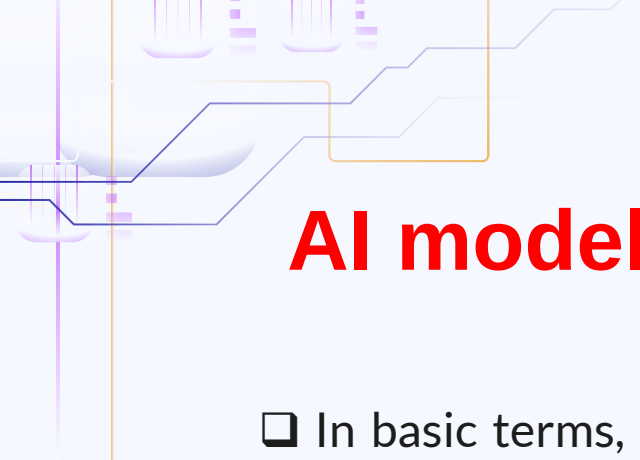
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50%	3.000000	117.000000	72.000000	23.000000	30.500000	32.000000	0.372500	29.000000	0.000000
75%	6.000000	140.250000	80.000000	32.000000	127.250000	36.600000	0.626250	41.000000	1.000000
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AI Workflow



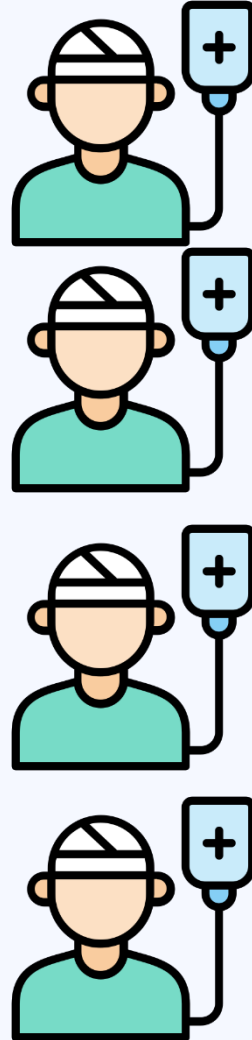


AI model

- ❑ In basic terms, ML is the process of training a piece of software, called a **model**, to make useful predictions from data.
- ❑ For example, suppose we wanted to create an app to predict rainfall.
 - ❑ Using an ML approach, we would give an ML model enormous amounts of patient health data (such as blood pressure, cholesterol, age, and ECG results) until the ML model eventually learned the mathematical relationship between clinical metrics and heart disease risk (**Training**).
 - ❑ We would then give the model a new patient's current health data, and it would predict their risk of heart disease (**Testing**).

AI Model

Training



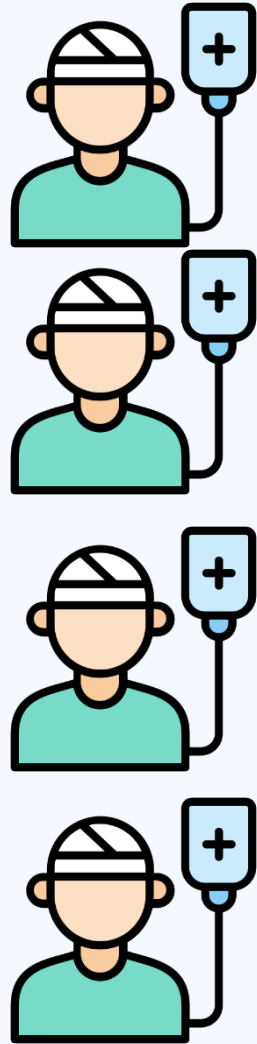
Testing



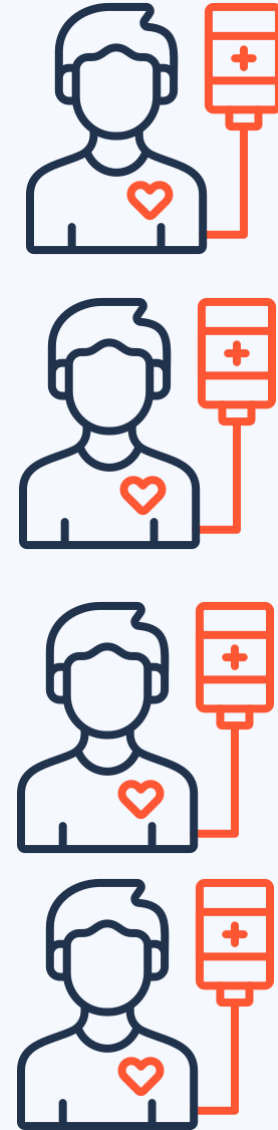
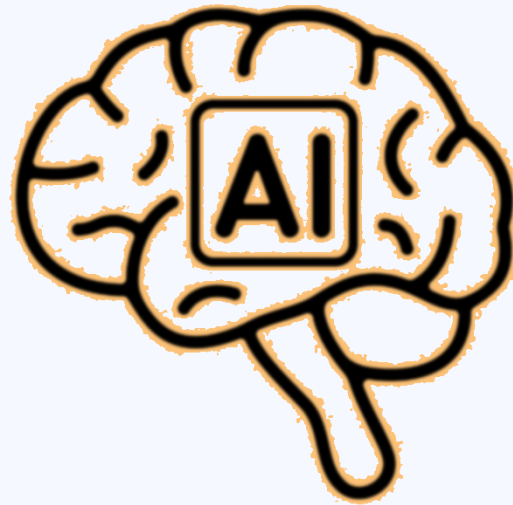
Training Patients on a Specific Disease

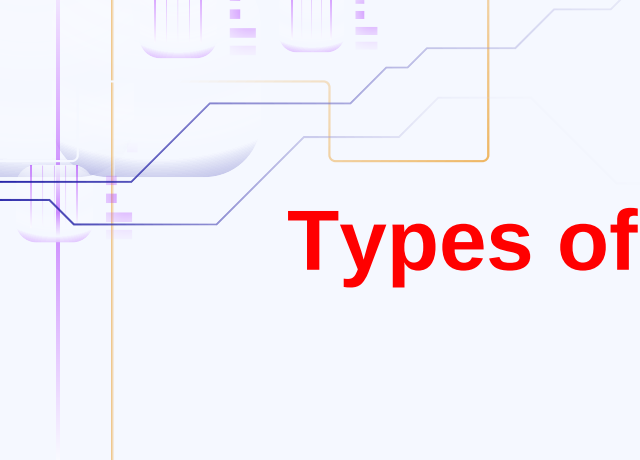
Real-life Patients on your Clinic

Training Data



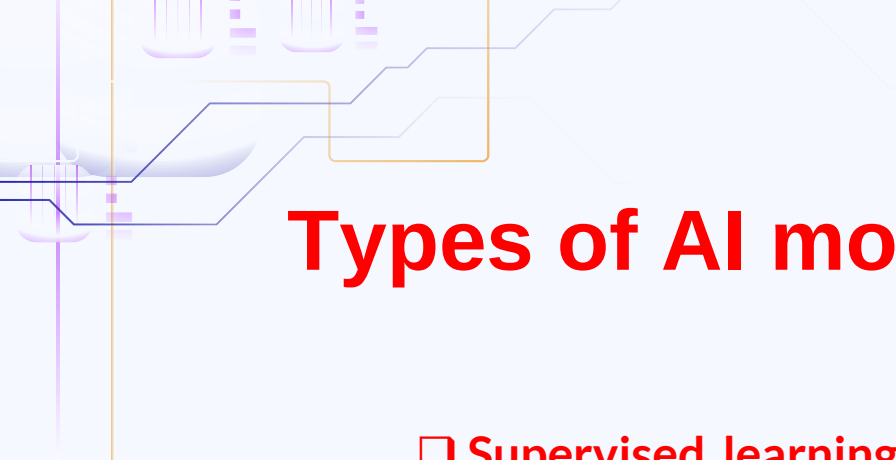
Testing Data





Types of AI models

- ☐ Supervised learning
- ☐ Unsupervised learning
- ☐ Reinforcement learning
- ☐ Generative AI



Types of AI models

❑ **Supervised learning:** can make predictions after seeing lots of data with the correct answers and then discovering the connections between the elements in the data that produce the correct answers. Two of the most common use cases for supervised learning are **regression** and **classification**.

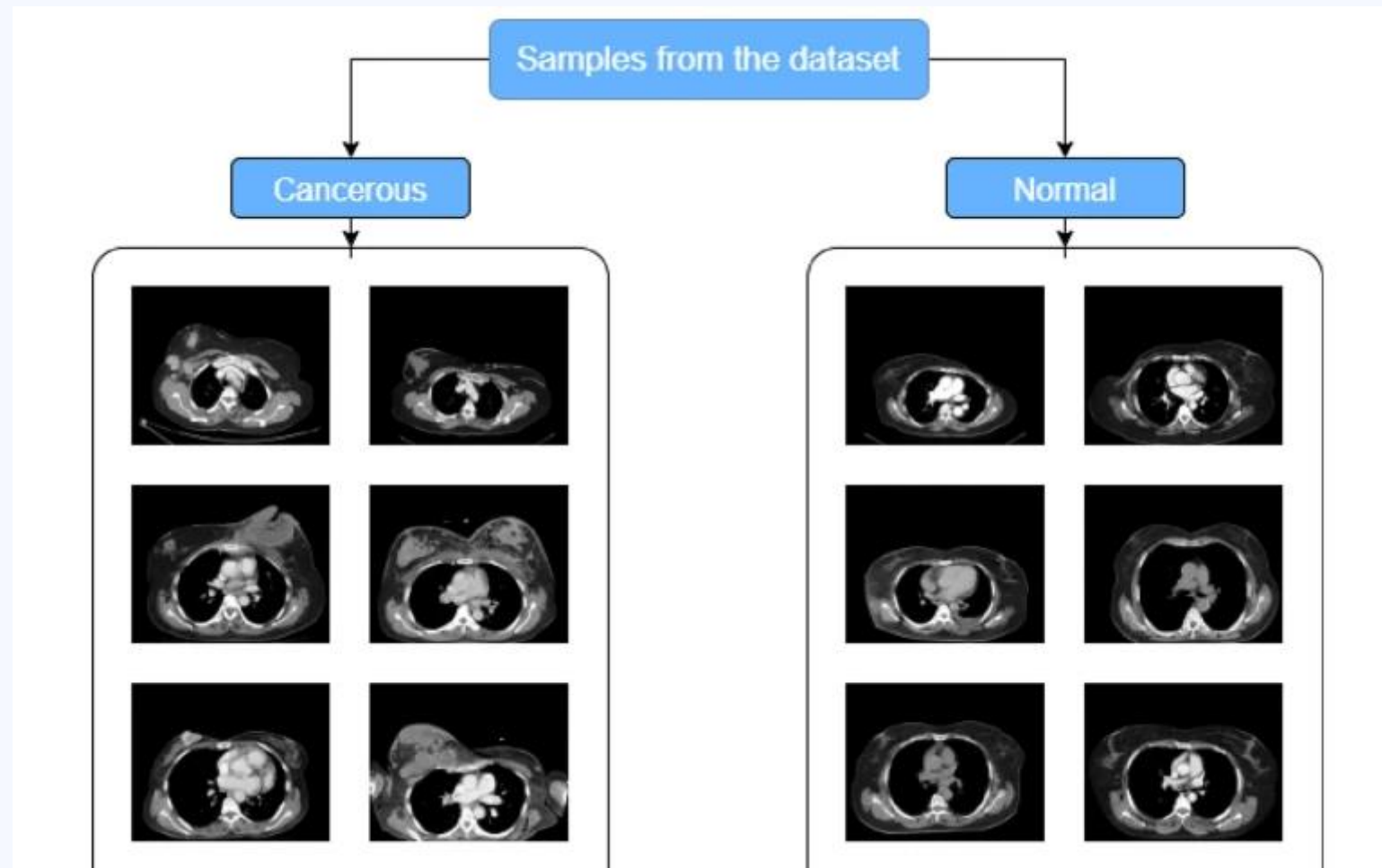
❑ **A regression model** predicts **a numeric value**. For example a medical model that predicts:

- a patient's blood glucose level, in mg/dL
- a patient's systolic blood pressure, in mmHg
- predicts a patient's length of hospital stay, in days
- predicts the size of a tumor, in millimeters

Types of AI models

- ☐ Supervised learning
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- ☐ Generative AI

Breast Cancer CT-Scan Image Classification



Class/Label= Cancerous

Class/Label= Normal

Types of AI models

FIGURE 1:

SIMPLIFIED INDIVIDUAL PATIENT GLUCOSE PREDICTIONS

Track A:
Adult Female,
Active Life



INPUT DATA:



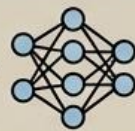
Age: 45



BMI: 24



Meal: 2 hrs ago
(small snack)



ML PREDICTION MODEL
(SINGLE TIME-POINT
CALCULATION)



**Predicted
Glucose:**
110 mg/dL
(Single numeric value
for this moment)

Track B:
Senior Male,
Sedentary Life



INPUT DATA:



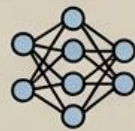
Age: 72



BMI: 31



Meal: 4 hrs ago
(full meal)



ML PREDICTION MODEL
(SINGLE TIME-POINT
CALCULATION)

**Predicted
Glucose:**
165 mg/dL
(Single numeric value
for this moment)

Track C:
Child Male,
High Activity



INPUT DATA:



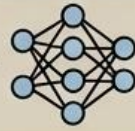
Age: 10



BMI: 18



Meal: 1 hr ago
(lunch)

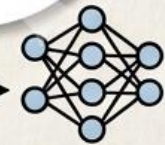


ML PREDICTION MODEL
(SINGLE TIME-POINT
CALCULATION)



**Predicted
Glucose:**
95 mg/dL
(Single numeric value
for this moment)

EXAMPLE: An ML model is applied to current data for three different patients at a single, specific time-point. The model takes their individual input features and calculates a unique, precise numeric prediction for their glucose level.





Classification AI models

❑ **Classification models** predict the likelihood that something belongs to a category.

❑ Classification models output a value that states whether or not something belongs to a particular category. For example, classification models are used to predict if a patient has cancer or not.

❑ Classification models are divided into two groups: **binary classification** and **multiclass classification**.

❑ **Binary classification** models output a value from a class that contains only two values, for example, a model that outputs either rain or no rain.

❑ **Multiclass classification** models output a value from a class that contains more than two values, for example, a model that can output either breast, colon, oral cancers.

Classification AI models

❑ **Example:** using gene expression data, you can predict different types of cancer.

	gene_1	gene_2	gene_3	gene_4	gene_7999	gene_8000	Cancer_Type
0	0.0	0.000000	0.0	2.088413	6.928584	2.088413	KIRC
1	0.0	0.000000	0.0	3.205955	5.039231	2.448002	KIRC
2	0.0	0.000000	0.0	4.746646	3.119738	4.433988	BRCA
3	0.0	0.000000	0.0	1.173191	2.863046	3.380342	KIRC
4	0.0	0.000000	0.0	1.366532	6.544487	0.839395	COAD
...	...	...	...	...	...	...	...
796	0.0	2.933384	0.0	1.683921	6.396848	1.252839	BRCA
797	0.0	0.000000	0.0	2.772625	5.386625	1.697640	COAD
798	0.0	0.408277	0.0	1.854754	5.507417	1.398131	BRCA
799	0.0	0.639232	0.0	3.226971	5.282158	0.639232	BRCA
800	0.0	0.000000	0.0	2.336912	7.550000	0.926379	COAD

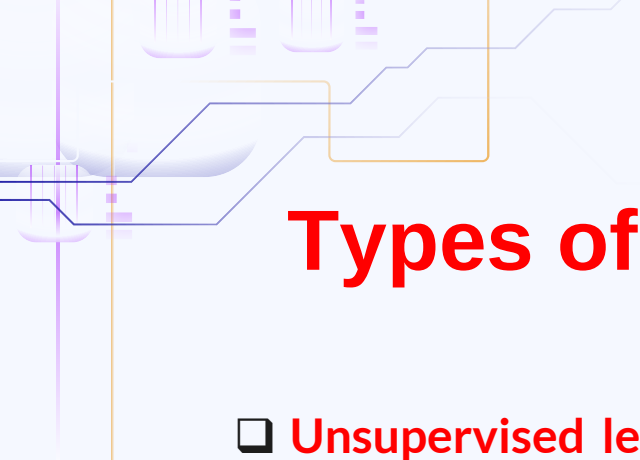
Kidney renal clear cell carcinoma (**KIRC**)

Breast invasive carcinoma (**BRCA**)

Colon adenocarcinoma (**COAD**)

Lung adenocarcinoma (**LUAD**)

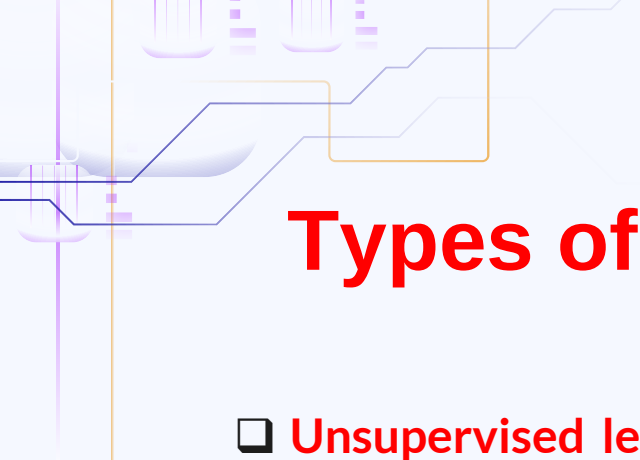
Prostate adenocarcinoma (**PRAD**)



Types of AI models

❑ **Unsupervised learning models** make predictions by being given data that does not contain any correct answers. An unsupervised learning model's goal is to identify meaningful patterns among the data. In other words, the model has no hints on how to categorize each piece of data, but instead it must infer its own rules.

❑ **For example**, an unsupervised model might cluster a patient dataset based on vital signs and lab results, revealing segmentations that define distinct risk profiles. You might then attempt to name those clusters (such as "**Healthy**," "**Pre-Diabetic**," or "**High Risk**") based on your understanding of the dataset.

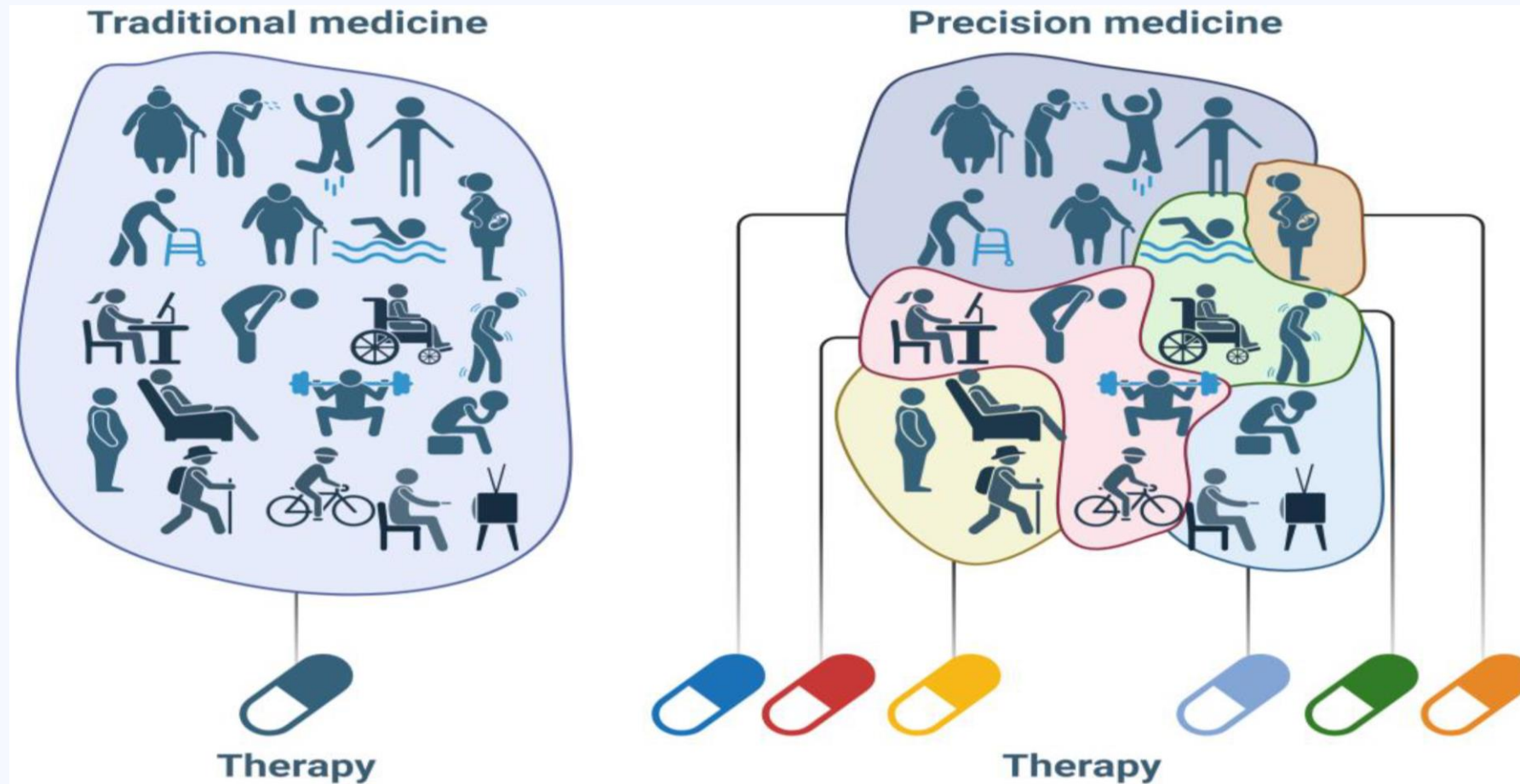


Types of AI models

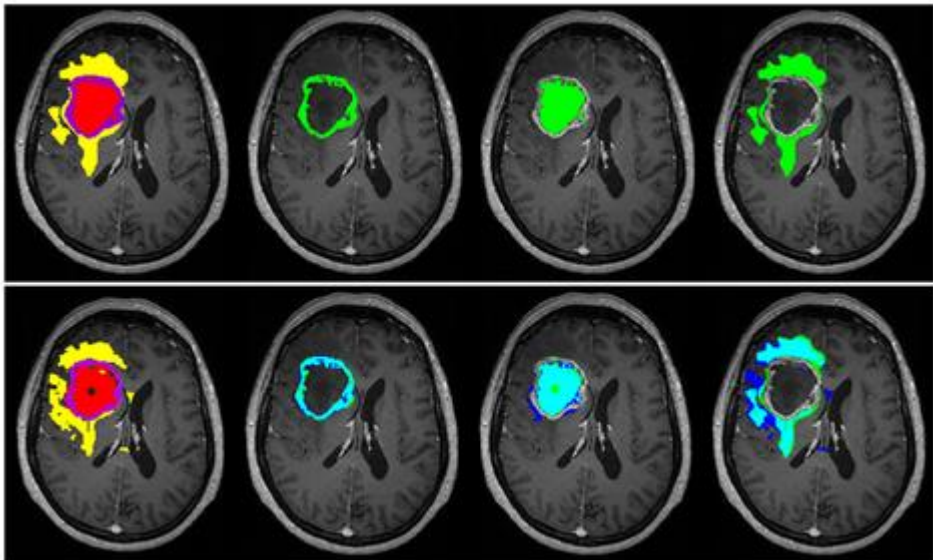
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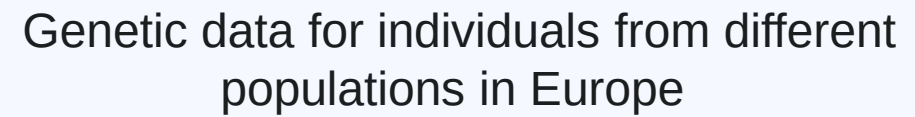
Types of AI models

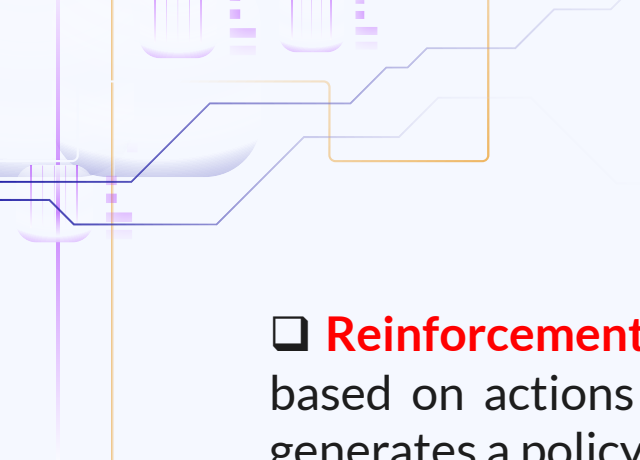


Types of AI models



The image displays two rows of four brain MRI slices each, illustrating the output of different AI models for tumor segmentation. The top row shows results from four different models, and the bottom row shows results from another set of four models. The segmentation colors vary between models, indicating different levels of detail and accuracy in identifying the tumor regions.





Types of AI models

❑ **Reinforcement learning** models make predictions by getting rewards or penalties based on actions performed within an environment. A reinforcement learning system generates a policy that defines the best strategy for getting the most rewards.

❑ For example, a reinforcement learning model can learn to optimize insulin dosing for diabetic patients.

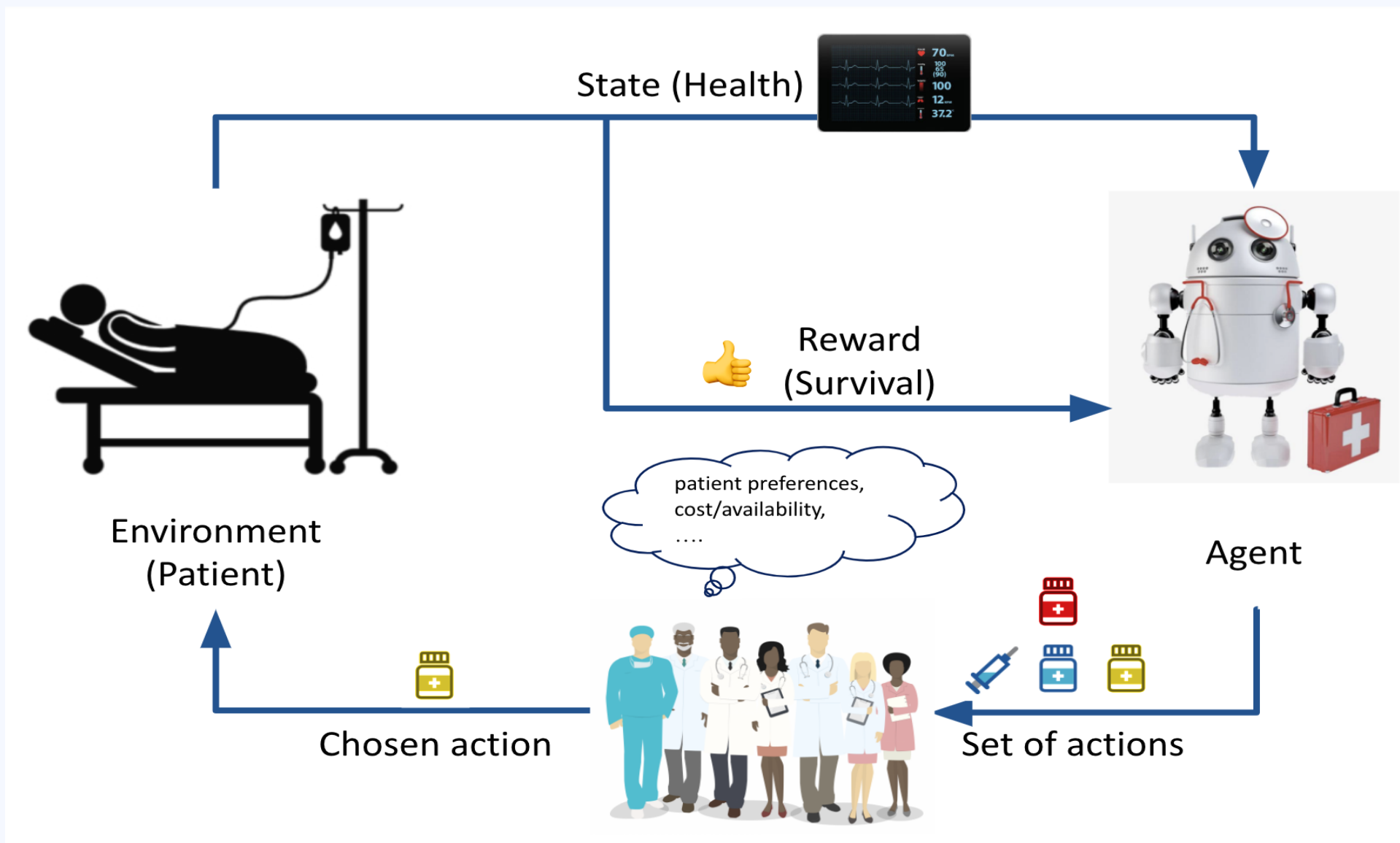
- The AI agent observes the patient's current blood glucose level and recent meal data (**State**).
- It decides on an optimal insulin dose to deliver (**Action**).
- It receives positive feedback if blood glucose normalizes, or a penalty if sugar spikes too high or drops too low (**Reward**), continually improving its treatment policy through trial and error over time.

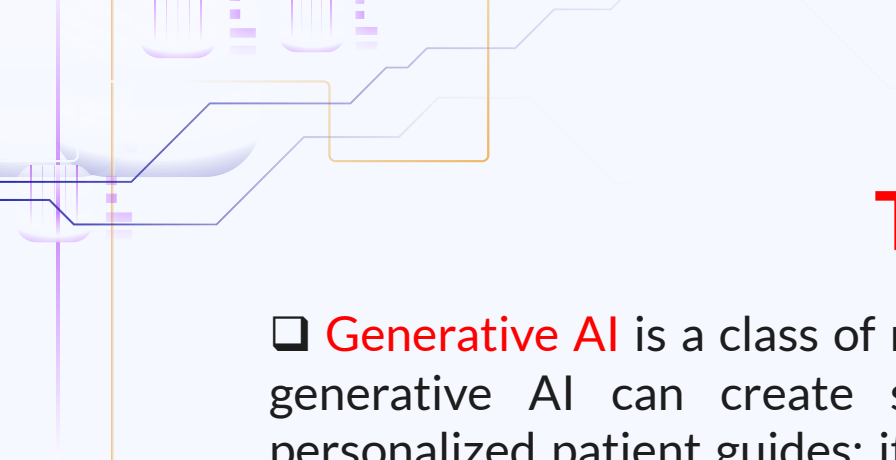
How Reinforcement Learning Maps to Medicine

❑ Unlike Supervised Learning (which learns from labeled historical data) or Unsupervised Learning (which finds hidden patterns), Reinforcement Learning learns by making decisions and receiving feedback (rewards/penalties).

RL Component	Medical Concept Example (Sepsis / ICU Treatment)
Agent	The AI clinical decision support system
Environment	The patient's body in the Intensive Care Unit (ICU)
State (S)	Patient's current vital signs (heart rate, blood pressure, oxygen, lab results)
Action (A)	Administering a specific dosage of medication or fluid adjustment
Reward (R)	Positive: Stabilization of organ function / patient recovery Negative (Penalty): Sudden drop in blood pressure or adverse side effects

How Reinforcement Learning Maps to Medicine





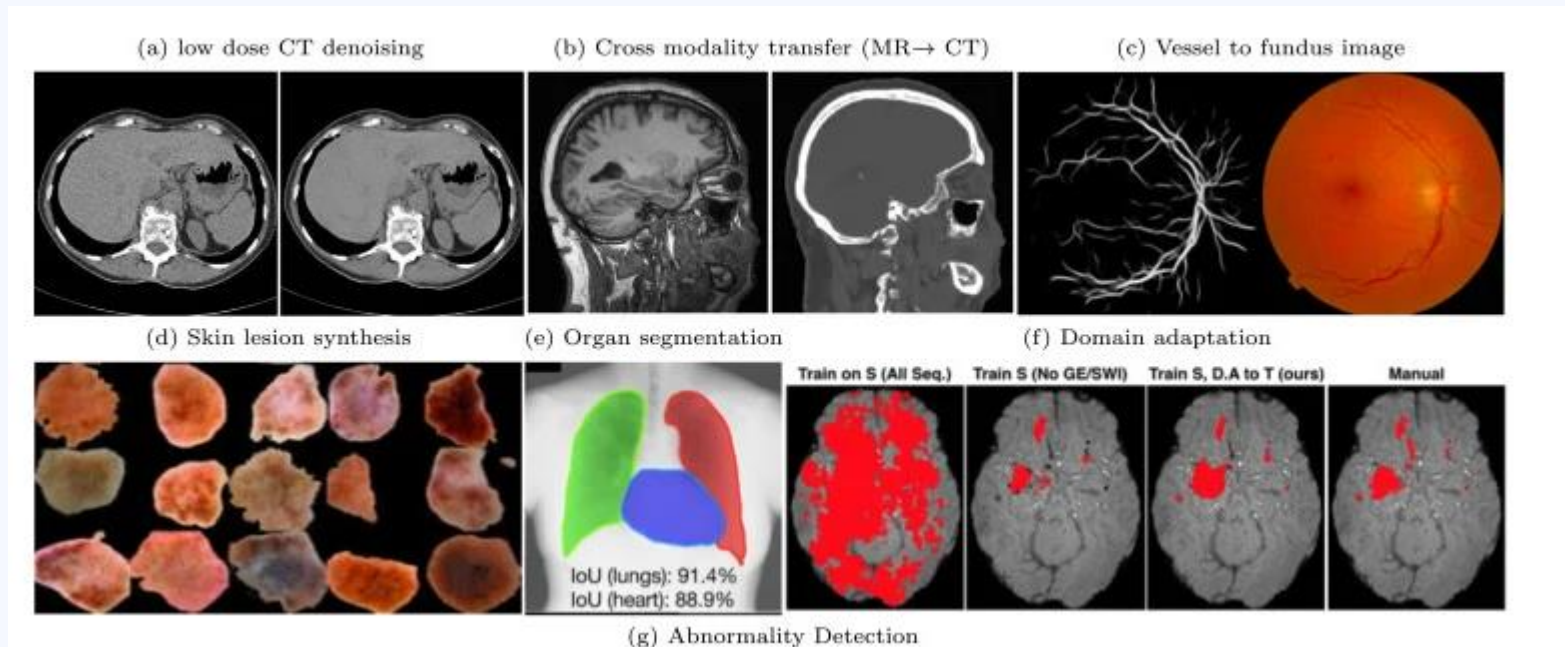
Types of AI models

❑ **Generative AI** is a class of models that creates content from user input. For example, generative AI can create synthetic medical images, novel drug molecules, and personalized patient guides; it can summarize clinical notes, explain treatment plans, or enhance radiology scans.

❑ **Generative AI** can take a variety of inputs and create a variety of outputs, like text, images, audio, and video. It can also take and create combinations of these. For example, a model can take an X-ray image as input and create an annotated scan and diagnostic report as output, or take a CT scan and physician notes as input and create a 3D surgical simulation video as output.

Types of AI models

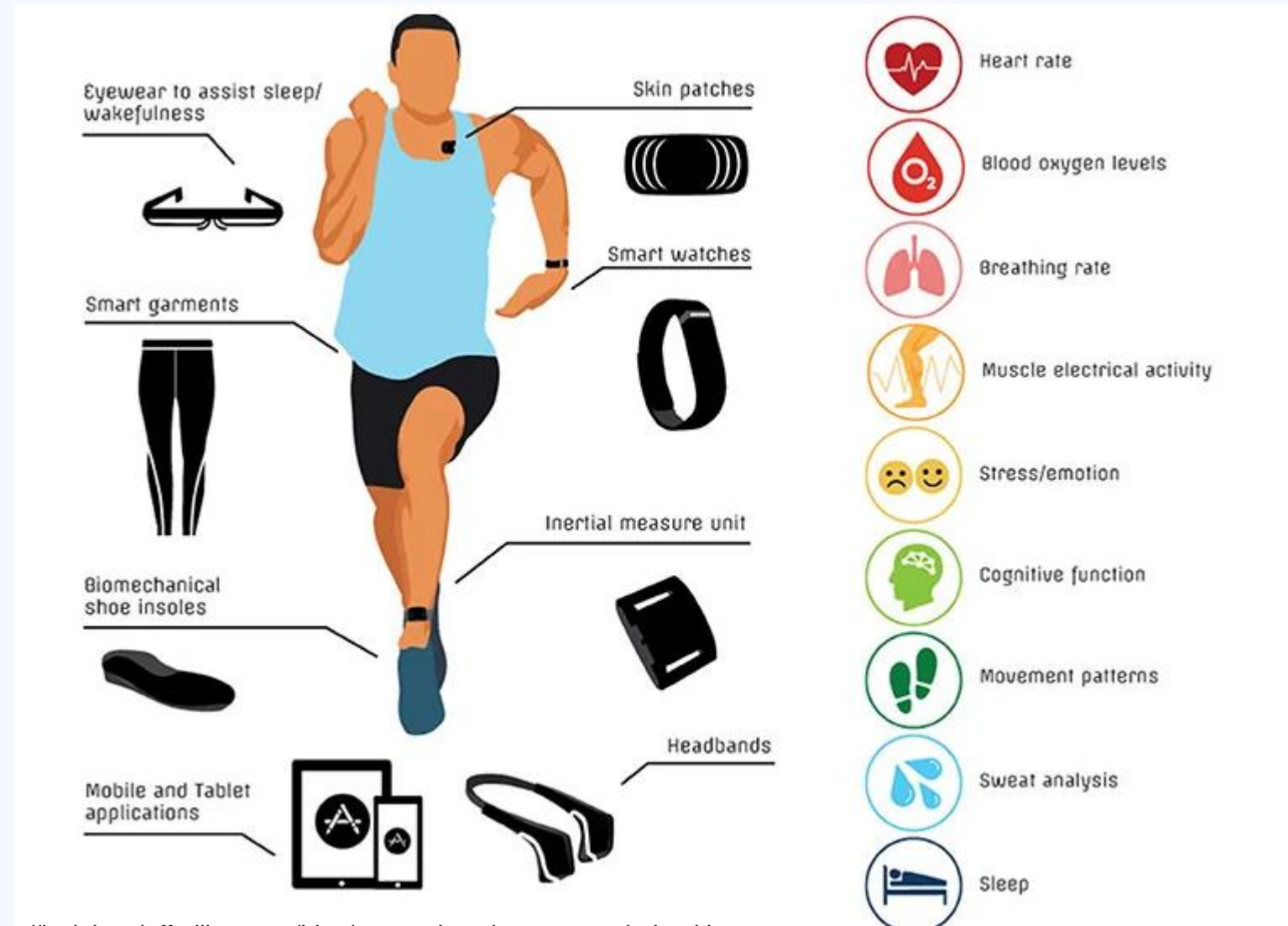
Generative AI can enhance image clarity by reducing background noises



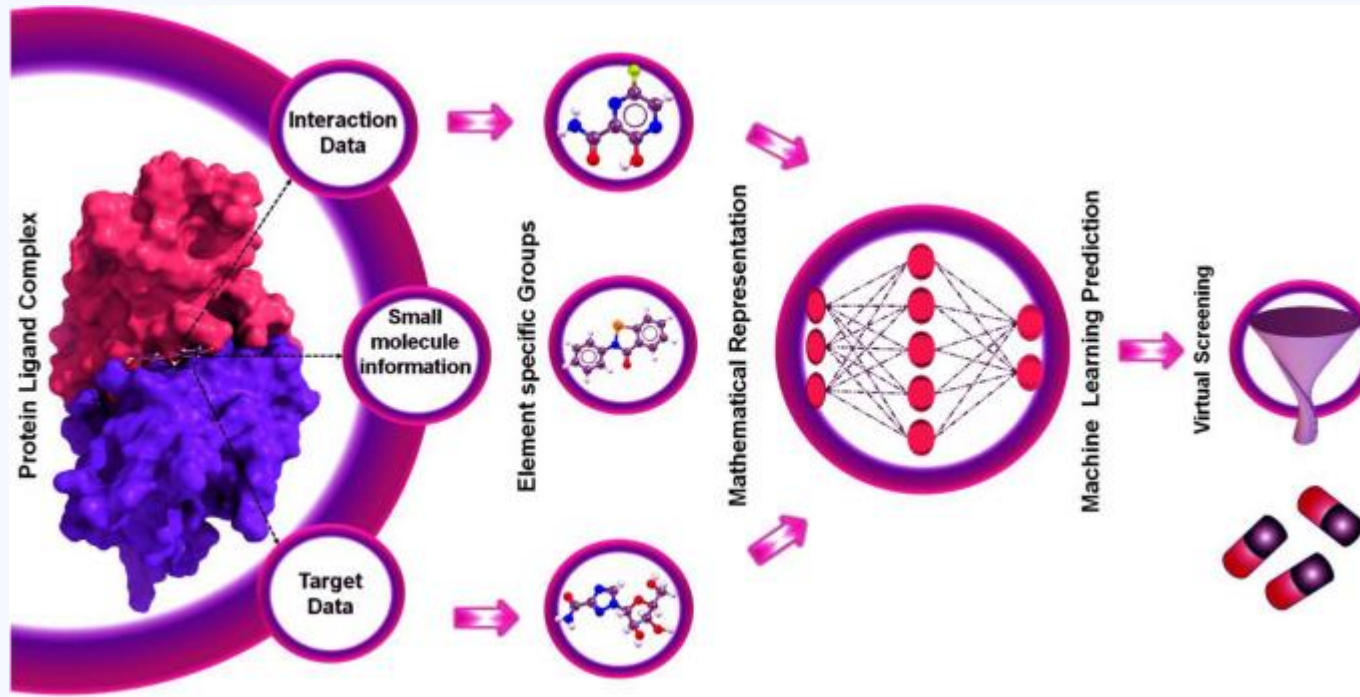
<https://insights.daffodilsw.com/blog/generative-ai-use-cases-in-healthcare>

Types of AI models

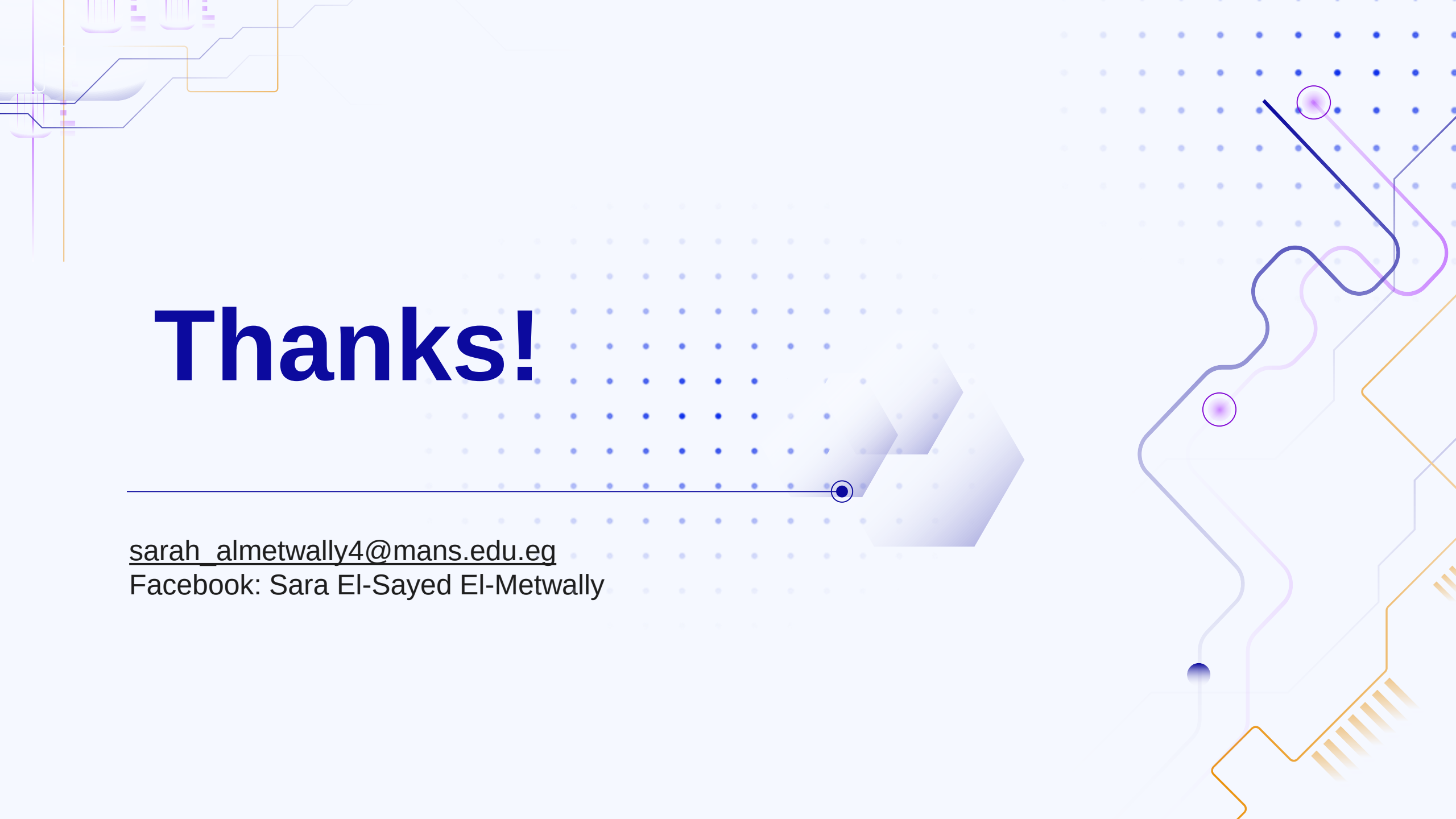
Companies developing medical devices are utilizing **GenAI** to design smart tools that assist in patient care



Types of AI models



Generative AI can create virtual molecules. By leveraging GenAI, pharmaceutical scientists can develop these virtual compounds and evaluate them using computer simulations instead of conducting physical experiments.



Thanks!

sarah_almetwally4@mans.edu.eg

Facebook: Sara El-Sayed El-Metwally